

Fundamental sources of THz radiation such as backward-wave oscillators, free-electron and quantum cascade lasers, novel heterojunction diode and transistor circuits, and—most recently—micromachined vacuum electronic devices are now beginning to round out the THz toolset.

However, a widely tunable THz-wave source has long been desired in the frequency region above 1 THz, where the tuning capability of a backward-wave oscillator (BWO) rapidly decreases. Widely tunable sources covering the region between 1 and 3 THz are limited to free electron lasers or photomixers. In photomixers, two optical lasers are mixed to generate tunable THz waves. This should become a reliable source if the limitation on the output due to the non-destructive limit of a photomixer is overcome.

Micro-fabricated vacuum electron devices represent a promising solution for obtaining high-power (>1mW CW) compact sources of the THz radiation. Typical fast-wave vacuum electronic devices in this regime—such as gyrotrons and free-electron lasers—tend to be large, expensive, high-voltage and very high-power and hence unsuitable for most of the applications aforementioned.

Narrow-band THz radiation can be produced by free-electron lasers and fast diodes.

Broadband THz radiation can be produced by thermal sources and, more recently, by table-top laser-driven sources and by short electron bunches in accelerators, but so far only with low power.

Sources of THz radiation are, at this point, very rare in physics and materials science laboratories and almost non-existent in chemistry, biology and medical laboratories. The barriers to performing experiments using THz radiation are enormous.

Most of the THz sources cannot generate THz beam with greater power, although it has many uses. The concept of new-generation sources provides THz radiation that is rich in

potential and novel applications, using relativistic electron beam.

Extreme nonlinear quantum phenomena can be explored using THz electromagnetic fields without exceeding material damage thresholds. Such phenomena occur when the ponderomotive energy (kinetic energy of a conduction electron oscillating in the THz field) or electric dipole coupling strength (THz electric field times dipole moment of a quantum-confined carrier) exceeds the photon energy. The photon energy at THz frequencies is large enough to enforce dynamic behaviour, but small enough to enter the extreme nonlinear quantum regime with reasonable electric fields.

In the microwave region of the spectrum, it is difficult to fabricate electronic devices that operate at frequencies above a few hundred GHz. This is a consequence of only low powers being delivered by devices which must have small device areas to minimise the device capacitance.

In the optical region of the spectrum, interband diode lasers have long been designated for operation at visible and near infrared frequencies. However, this concept where light is generated by the radiative recombination of conduction-band electrons and valance-band holes across the band gap of the active material cannot be simply extended into the mid infrared since suitable semiconductors are not available.

Recent research has made extensive use of the quantum cascade concept in which inter sub-band or inter miniband transitions in layered semiconductor heterostructures are used to generate light in the mid infrared region of the spectrum. Owing to the difficulties in fabricating solid-state THz sources, researchers have focused attention on all-optical techniques of producing THz radiation employing visible/near-infrared, femtosecond pulsed lasers.

A p-type germanium laser is another widely tunable source in this region. Solid-state electronics, vacuum

electronics, microwave techniques, ultrafast visible and near-infrared (NIR) lasers, single-mode continuous-wave NIR lasers, electron accelerators ranging in size from a few centimetres to a 1.6km-long linear accelerator, and novel materials have been combined to yield a large variety of sources with widely-varying output characteristics.

Conventional microwave sources at high frequencies do not work fast enough to efficiently produce radiation in the THz regime, whereas laser diode sources have been limited by thermal effects. A new approach called plasma wave electronics holds promise of developing tunable sources and tunable sensitive detectors of THz radiations. With an exception of the free electron lasers that use relativistic electron beams and are capable of reaching kW-level THz power, other sources generate mW or microwatt levels.

Future outlook

Difficulties in the development of THz sources have not stopped an explosion of interest in the THz technology. The reason is its emerging applications in imaging, medicine, biology, space exploration, covert communications, compact radar ranges, industrial control, THz microscopy, THz tomography and homeland security.

Recently developed practical sources and detectors of THz power promise to greatly expand the realm of potential applications of the THz frequency band. Further advances in coherent generation, amplification, antennae and modulation will make it possible to develop an array of electronic sources and detectors and offer a new spectroscopic imaging and screening technology. ●

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